



AI-Powered News Categorization, Summarization, and Multilingual Recommendation System Using Natural Language Processing

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

DSA0307 – Natural Language Processing

to the award of the degree of

**BACHELOR OF TECHNOLOGY IN
ARTIFICIAL INTELLIGENCE AND DATA SCIENCES**

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "AI-Powered News Categorization, Summarization, and Multilingual Recommendation System Using Natural Language Processing " has been carried out by “C.Nithisha” (192424375) under the supervision of Dr.Anbazhagan. K is submitted in partial fulfilment of the requirements for the current semester of the B.Tech (Artificial Intelligence and Data sciences) programme at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I C Nithisha (192424375) of the Department of Artificial Intelligence and Data Science, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled " AI-Powered News Categorization, Summarization, and Multilingual Recommendation System Using Natural Language Processing " is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Hashini S	Dataset management, system integration, testing, documentation	Integrated the three modules, managed database/dataset operations, connected frontend and backend, and supported deployment	End-to-end system testing, test-case preparation, result analysis, bug identification and correction	Methodology, Implementation, Testing & Results, Appendices, supporting documentation	30%
Roshini B T	Dashboard/UI design, recommendation engine, deployment, documentation	Built the personalized recommendation logic and progress-dashboard module (Module 3)	Usability testing, system integration testing	Chapters 4.6–4.8, 5, Abstract, References	35%
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ABSTRACT

The rapid growth of digital news platforms has made it difficult for users to efficiently identify relevant, reliable, and understandable news from the large volume of information available online. News articles are continuously generated across different categories and languages, making manual classification, summarization, and personalized news selection time-consuming. This project addresses the engineering problem of developing an integrated Natural Language Processing (NLP)-based system that can automatically organize news content and provide concise and personalized information to users.

The proposed system, titled “AI-Powered News Categorization, Summarization, and Multilingual Recommendation System Using Natural Language Processing,” combines four major modules: news data collection and text preprocessing, AI-based news categorization, automatic news summarization, and multilingual news recommendation. In the first module, news articles are collected and processed using NLP techniques such as text cleaning, tokenization, stop-word removal, and lemmatization. The processed text is then used by the categorization module to classify news into relevant categories such as Politics, Sports, Technology, Business, and Entertainment. The summarization module identifies important information from an article and generates a shorter representation of the original content. Finally, the recommendation module uses category, language, and content similarity to provide relevant news to users in a multilingual format.

The system is designed and implemented using Python and appropriate NLP and machine learning libraries. The performance of the classification and recommendation components is evaluated using suitable performance measures, while the generated summaries are examined for relevance and information preservation. The experimental results obtained from the implemented system are presented and analysed in Chapter 4.

The proposed system provides an integrated approach for reducing information overload and improving the accessibility of digital news. It demonstrates how NLP and machine learning techniques can be combined to automate multiple stages of news processing and deliver more relevant and concise information to users.

Keywords:

Natural Language Processing, News Categorization, Text Summarization, Multilingual Recommendation, Machine Learning, News Processing.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
ML	Machine Learning	—
IRT	Item Response Theory	—
MAE	Mean Absolute Error	marks
RMSE	Root Mean Squared Error	marks
API	Application Programming Interface	—
UI/UX	User Interface / User Experience	—
SO	Student Outcome	—

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

The rapid expansion of digital media and online news platforms has significantly changed the way people access information. News is available through websites, mobile applications, social media platforms, and other digital sources. Although this provides users with quick access to information, the large amount of continuously generated news content creates an information overload problem. Users may have difficulty identifying articles that are relevant to their interests and understanding lengthy articles within a limited amount of time.

Natural Language Processing (NLP) provides techniques for enabling computers to process and understand human language. NLP techniques can be applied to news articles for tasks such as text preprocessing, classification, summarization, translation, and recommendation. Machine learning methods can further automate the process of identifying patterns in news content and assigning appropriate categories.

The proposed project integrates these capabilities into a single system for intelligent news processing. The system consists of four major modules: News Data Collection and Text Preprocessing, AI-Based News Categorization, Automatic News Summarization, and Multilingual News Recommendation. These modules work together to transform raw news articles into categorized, summarized, and personalized multilingual news information.

1.2 Need for the Project

The increasing number of online news articles makes it difficult for users to manually read and organize all available information. Different users may also have different interests, such as technology, sports, politics, business, or entertainment. Therefore, a system that can automatically understand news content and present relevant information can reduce the effort required by users.

The major needs addressed by the project are:

- To reduce the time required to identify relevant news.

- To automatically categorize large numbers of news articles.
- To provide concise summaries of lengthy news articles.
- To improve accessibility for users who prefer different languages.
- To provide personalized recommendations based on news content and user preferences.
- To integrate multiple NLP-based news processing tasks into a single system.

From an engineering perspective, the project involves multiple interacting components including data processing, NLP, machine learning, summarization, language processing, and recommendation. Therefore, an integrated solution is required rather than treating each task independently.

1.3 Problem Statement

Digital news platforms generate a large and continuously increasing volume of textual information. Manually identifying the category of each article, reading complete articles, generating summaries, and finding relevant news in a preferred language is inefficient and time-consuming.

The engineering problem is therefore to design and develop an NLP-based intelligent news processing system that can automatically preprocess news articles, classify them into appropriate categories, generate concise summaries, and recommend relevant news in multiple languages while maintaining useful information from the original content.

The system must handle variations in textual content, category differences, language requirements, and user preferences while providing useful and relevant output.

1.4 Project Aim

The main aim of this project is to design and develop an AI-powered NLP system for automatic news categorization, summarization, and multilingual recommendation to improve the efficiency, relevance, and accessibility of digital news consumption.

1.5 Project Objectives

The objectives of the proposed project are:

1. To collect and prepare news article data for NLP-based processing.
2. To develop a text preprocessing pipeline involving cleaning, tokenization, stop-word removal, and lemmatization.
3. To develop a machine learning/NLP-based model for automatically categorizing news articles.
4. To implement an automatic summarization component for generating concise representations of news articles.
5. To develop a multilingual recommendation component for providing relevant news based on content, category, language, and user preferences.
6. To integrate the four modules into a unified news processing system.
7. To test and evaluate the developed system using suitable performance measures.
8. To analyse the strengths, limitations, and possible improvements of the proposed system.

1.6 Scope

The scope of the project covers the development of an NLP-based software system for processing digital news articles.

Included in the Scope

- Collection or loading of news article datasets.
- Text cleaning and preprocessing.
- Tokenization and removal of unnecessary words.
- Lemmatization of textual content.
- Extraction of useful textual features.
- Automatic news category classification.
- Automatic generation of concise news summaries.

- Multilingual processing and recommendation.
- Content-based recommendation of relevant news.
- Evaluation of classification and recommendation performance.
- Integration of the individual modules into one system.

Excluded from the Scope

- Verification of the factual correctness of every news article.
- Detection of completely fabricated or fake news unless specifically implemented as an additional component.
- Professional-level human translation for every possible language.
- Replacement of professional journalists or editors.
- Real-time processing of every news source available on the internet.

Assumptions

The system assumes that the input news articles contain sufficient textual information for NLP processing. It also assumes that the available dataset contains appropriate category information when supervised learning is used for classification.

Boundary Conditions

The performance of the system depends on the quality and size of the available dataset, the selected NLP techniques, the trained model, language resources, and computational resources. The recommendation quality may also vary depending on the amount of available user preference or interaction information.

1.7 Expected Engineering Outcome

The expected outcome of the project is a functional AI-powered news processing prototype capable of performing four major tasks:

Input News → Text Preprocessing → News Categorization → Summarization → Multilingual Recommendation → User Output

The developed system is expected to automatically organize news articles into suitable categories, generate shorter versions of lengthy articles, and recommend relevant content according to user interests and language preferences.

The project is also expected to demonstrate the practical application of NLP and machine learning techniques in solving an information overload problem. The final system will provide an integrated software-based solution that can be further extended with real-time news APIs, additional languages, advanced recommendation algorithms, and improved summarization models.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

A literature review was carried out to understand the existing techniques and systems used for automatic news processing. The review mainly focused on Natural Language Processing (NLP), text preprocessing, news classification, automatic text summarization, and multilingual recommendation systems.

Existing research and solutions were studied based on their methods, advantages, limitations, and suitability for developing an integrated news processing system. The review considered machine learning and NLP-based approaches used for classifying news articles into different categories. Different summarization techniques were also analysed to understand how lengthy news articles can be converted into shorter and meaningful summaries.

In addition, multilingual NLP approaches were reviewed to understand how news content can be processed and recommended for users with different language preferences. The review helped identify suitable techniques that can be combined to develop the proposed system.

The major areas considered during the review are:

1. NLP-based text preprocessing.
2. Machine learning-based news categorization.
3. Automatic text summarization.
4. Multilingual text processing.
5. Content-based news recommendation.
6. Integration of multiple NLP functions into a single system.

2.2 Review of Existing Work

2.2.1 Natural Language Processing for News Processing

Natural Language Processing is widely used for processing and understanding human language. In news applications, NLP techniques are used to convert unstructured article text into a form that can be processed by machine learning algorithms.

Common preprocessing operations include text cleaning, tokenization, removal of stop words, stemming or lemmatization, and feature extraction. These operations reduce unnecessary information and improve the quality of textual data used for further processing.

For news articles, preprocessing is important because articles may contain punctuation, special characters, repeated words, and other unnecessary information. A proper preprocessing pipeline can improve the performance of classification and recommendation systems.

2.2.2 News Categorization

News categorization is the process of assigning a news article to a predefined category. Common categories include Politics, Sports, Business, Technology, Entertainment, and World News.

Machine learning algorithms can be trained using labelled news datasets to automatically classify new articles. Traditional approaches commonly use text features such as Term Frequency-Inverse Document Frequency (TF-IDF). Classification algorithms such as Naive Bayes, Logistic Regression, Support Vector Machine, and Decision Tree can then be applied to the extracted features.

The main advantage of automatic classification is that it reduces the need for manual organization of large numbers of news articles.

2.2.3 Automatic News Summarization

News summarization aims to reduce a lengthy article into a shorter version while preserving its important information. Summarization techniques are generally classified into extractive and abstractive approaches.

Extractive summarization selects important sentences or phrases directly from the original article. It is relatively simple to implement and generally preserves the original wording.

Abstractive summarization generates new sentences that represent the main meaning of the original article. It can produce more natural summaries but generally requires more advanced NLP models and computational resources.

For the proposed project, summarization is included as an important component for reducing the time required by users to understand lengthy news articles.

2.2.4 Multilingual News Processing

Digital news is available in multiple languages. A multilingual news system can improve accessibility by allowing users to receive or process information according to their preferred language.

Multilingual NLP can involve language identification, translation, multilingual embeddings, and language-specific preprocessing. However, different languages have different grammatical structures, vocabularies, and available language resources.

Therefore, multilingual recommendation requires consideration of both the content of the article and the preferred language of the user.

2.2.5 News Recommendation

News recommendation systems are designed to provide users with articles that are relevant to their interests. Content-based recommendation is one commonly used approach. It recommends articles by comparing the characteristics or textual content of articles with the user's interests or previously selected content.

Similarity measures can be used to identify articles that are closely related to a user's preferred topics. Combining category information, content similarity, and language preference can improve the relevance of recommended news.

2.3 Comparative Analysis

Existing approaches for news processing were compared based on their functionality and limitations.

Table 2.1: Comparison of Different Approaches for News Classification and Processing

Approach	Main Function	Advantages	Limitations
Manual News Classification	Organizes news manually	Simple to understand	Time-consuming and difficult for large datasets
Machine Learning Classification	Automatically classifies news	Fast and scalable	Requires suitable training data
Extractive Summarization	Selects important sentences	Simple and preserves original content	May not produce natural summaries
Abstractive Summarization	Generates new summaries	More flexible and natural	Requires more computational resources
Content-Based Recommendation	Recommends similar articles	Personalised based on content	May have limited diversity
Multilingual Processing	Handles multiple languages	Improves accessibility	Language resources and translation quality may vary
Integrated NLP System	Combines multiple functions	Provides complete news-processing workflow	More complex to design and integrate

The comparison shows that individual approaches solve specific problems effectively, but an integrated system can provide a more complete solution by combining categorization, summarization, and recommendation.

2.4 Research / Engineering Gap

From the review of existing approaches, it was observed that news categorization, summarization, and recommendation are often treated as separate tasks. A system that combines these functions can provide a more convenient workflow for users.

The major engineering gaps identified are:

1. **Lack of integration:** Many approaches focus on only one task, such as classification or summarization.
2. **Information overload:** Users may still need to read large amounts of content even after articles are categorized.
3. **Limited personalization:** Basic news systems may not sufficiently consider user interests and language preferences.
4. **Multilingual challenges:** Processing news across multiple languages requires additional language-specific handling.
5. **End-to-end processing:** There is a need for a unified workflow that processes news from collection to final recommendation.

Therefore, the proposed project aims to address these gaps by integrating multiple NLP-based functions into a single system.

2.5 Proposed Contribution

The system is intended to reduce information overload, improve news accessibility, and help users identify relevant information more efficiently. The implementation and experimental evaluation of the proposed system are presented in Chapter 4.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

The proposed system requires both functional and non-functional requirements to perform news collection, preprocessing, categorization, summarization, and multilingual recommendation effectively.

3.1.1 Functional Requirements

The major functional requirements of the system are:

1. The system should accept news articles as input.
2. The system should clean and preprocess the news text.
3. The system should tokenize the processed text.
4. The system should remove unnecessary stop words.
5. The system should perform lemmatization where required.
6. The system should classify news articles into predefined categories.
7. The system should generate a concise summary of the selected news article.
8. The system should identify or handle the language of the news content.
9. The system should recommend relevant news based on content, category, and language preference.
10. The system should display the processed news and recommendation results to the user.

3.1.2 Non-Functional Requirements

The system should satisfy the following non-functional requirements:

- **Accuracy:** The classification and recommendation results should be sufficiently accurate.
- **Efficiency:** The system should process news within a reasonable amount of time.
- **Usability:** The interface should be simple and easy to understand.
- **Scalability:** The system should allow additional news categories and languages to be added.
- **Maintainability:** Individual modules should be easy to modify and update.
- **Reliability:** The system should provide consistent results for valid inputs.

3.2 Constraints & Applicable Standards

3.2.1 Constraints

The development of the proposed system is subject to several constraints:

1. The quality of results depends on the quality and size of the news dataset.
2. Classification performance depends on the selected machine learning algorithm and available training data.
3. Multilingual processing may be affected by the availability of language-specific NLP resources.
4. Advanced summarization models may require higher computational resources.
5. Recommendation quality depends on the available user preference and news-content information.
6. The system may not identify factual errors or misinformation unless a separate fact-checking or fake-news detection module is implemented.

3.2.2 Applicable Standards and Good Practices

The system follows general software engineering and responsible AI practices. The implementation emphasizes:

- Proper handling of input data.
- Consistent data preprocessing.
- Secure handling of user-related information.
- Reproducible machine learning experiments.
- Appropriate evaluation of model performance.
- Responsible use of news content and datasets.
- Clear separation between automated recommendations and verified factual information.

3.3 Design Specifications

Table 3.1: System Performance Parameters and Requirements

Parameter	Target/Requirement	Unit	Priority
News categorization accuracy	≥ 90	%	High
Summarization ROUGE-L score	≥ 0.35	Score	High
Translation latency	≤ 2	s/article	Medium
Recommendation Top-5 precision	≥ 80	%	High
Dashboard page load time	≤ 3	s	Medium
Minimum supported languages	≥ 3	Languages	Medium

3.4 Alternative Solutions & Evaluation

Different techniques can be considered for implementing the major components of the proposed system.

3.4.1 News Categorization

Table 3.2: Comparison of News Classification Algorithms

Method	Advantages	Limitations
Naive Bayes	Simple and fast	May not handle complex relationships effectively
Logistic Regression	Good for text classification	Depends on suitable feature extraction
Support Vector Machine	Effective for high-dimensional text	Can require more computation for large datasets
Deep Learning	Can learn complex patterns	Requires more data and computational resources

A suitable classification algorithm can be selected based on accuracy, dataset size, processing time, and computational requirements.

3.4.2 Text Summarization

Table 3.3: Comparison of Text Summarization Methods

Method	Advantages	Limitations
Extractive Summarization	Simple and fast	Uses sentences from the original article

Abstractive Summarization	Can generate natural summaries	More computationally demanding
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For the initial system, an extractive approach can be considered because it is easier to implement and requires fewer computational resources.

3.4.3 News Recommendation

Table 3.4: Comparison of News Recommendation Methods

Method	Advantages	Limitations
Content-Based	Uses article content and user interests	May recommend similar types of articles repeatedly
Collaborative Filtering	Uses preferences of similar users	Requires sufficient user interaction data
Hybrid Recommendation	Combines multiple methods	More complex to implement

A content-based approach is suitable for the proposed system because recommendations can be generated using article content and user interests without requiring a large number of users.

3.5 Selected Design & Detailed Design

Based on the comparison of different approaches, a modular NLP and machine learning architecture is selected.

Step 1 – News Input

News articles are collected or provided as input to the system.

Step 2 – Text Preprocessing

The news text is cleaned using operations such as:

Cleaning → Tokenization → Stop-word Removal → Lemmatization

Step 3 – Feature Extraction

The processed text is converted into numerical features. Techniques such as TF-IDF can be used to represent the text for machine learning.

Step 4 – News Classification

The extracted features are passed to the trained classification model. The model predicts the appropriate news category.

Step 5 – News Summarization

The article is processed by the summarization module to produce a concise summary containing the important information.

Step 6 – Multilingual Processing

The system handles different languages and provides content according to the user's language preference.

Step 7 – News Recommendation

The recommendation module compares the user's interests with the available news content and identifies relevant articles.

Step 8 – Final Output

The system displays the categorized news, summary, and recommended multilingual news to the user.

3.6 Trade-off Analysis

The following trade-offs were considered during system design.

3.6.1 Accuracy vs. Computational Cost

Advanced machine learning and deep learning models can provide improved accuracy but may require more computational resources. Simpler models provide faster processing and are easier to implement.

3.6.2 Summary Quality vs. Processing Time

Abstractive summarization can generate more natural summaries but requires greater computational resources. Extractive summarization is faster and simpler but may produce less natural summaries.

3.6.3 Personalization vs. Data Requirement

Highly personalized recommendations require more information about user preferences. A content-based recommendation approach can provide useful recommendations with comparatively less user data.

3.6.4 Language Support vs. System Complexity

Supporting additional languages improves accessibility but increases the complexity of preprocessing, translation, and language-specific processing.

3.6.5 Integration vs. Maintainability

Combining multiple NLP functions provides a complete system but increases development complexity. A modular architecture is therefore used so that each module can be tested and maintained independently.

3.7 Sustainability, Ethics, Safety, Risk & Security

3.7.1 Sustainability

The proposed system is mainly software-based and does not require specialized physical hardware. It can be implemented using commonly available computing resources.

Efficient algorithms can reduce unnecessary computational requirements. The modular design also allows individual components to be improved without completely redesigning the system.

3.7.2 Ethical Considerations

News recommendation systems can influence the information users receive. Therefore, ethical considerations are important.

The system should:

- Avoid intentionally biased recommendations.
- Respect user privacy.
- Avoid unnecessary collection of personal information.
- Give appropriate credit to news sources and datasets.
- Clearly indicate that automatically generated summaries may require verification.
- Avoid manipulating users through deliberately biased recommendations.

3.7.3 Safety Considerations

Incorrect categorization or summarization may result in users receiving incomplete or misleading information. Therefore, automatically generated summaries and recommendations should not be treated as a replacement for verified news sources. Important information should be checked against reliable sources when necessary.

3.7.4 Risk Analysis

Table 3.5: Risk Analysis and Mitigation Strategies

Risk	Possible Impact	Mitigation
Poor-quality dataset	Reduced model accuracy	Use suitable and cleaned datasets
Incorrect classification	Wrong news category	Test and evaluate the classification model
Poor summary	Important information may be missed	Evaluate summary quality using suitable metrics

Translation errors	Incorrect multilingual output	Use suitable language-processing resources
Irrelevant recommendations	Poor user experience	Improve similarity and recommendation methods
Data privacy issues	User information may be exposed	Minimize and protect user data
Model bias	Unfair recommendations	Analyse dataset and model behaviour
System failure	News processing may stop	Perform regular testing and error handling

3.7.5 Security Considerations

Security is considered during data collection, storage, processing, and user interaction. User-related information should be collected only when necessary and should be protected from unauthorized access. Input data should also be validated before processing.

If external APIs are used, API keys and authentication credentials should not be directly exposed in the source code. They should be stored securely using suitable environment variables or configuration methods.

Overall, the proposed design balances accuracy, processing speed, usability, scalability, computational requirements, privacy, and responsible use of AI.

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project is developed using a modular approach. The system consists of four main modules: data preprocessing, news categorization, summarization, and multilingual recommendation.

Table 4.1: Software Resources and Their Purpose

Resource	Purpose
Python	Main programming language
Pandas	Data processing
NLTK	NLP preprocessing
Scikit-learn	Classification
TF-IDF	Feature extraction
Matplotlib	Visualization
Streamlit	User interface
Google Colab	Development and testing

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

The project is a purely software-based system; no fabrication or physical hardware was involved. Development used Python for the NLP and machine-learning pipeline, Pandas and NumPy for data handling, NLTK for text preprocessing, and scikit-learn for news categorization. TF-IDF was used for text feature extraction, while summarization and multilingual processing techniques were used for generating concise and language-specific news content. Streamlit was used to develop the interactive dashboard, and Git/GitHub was used for version control and project management.

Table 4.2 : Tools, Software, Equipment and Their Usage

Tool / Software / Equipment	Purpose	Stage Used
Python 3.x	NLP processing and system development	Throughout project
Pandas, NumPy	Data cleaning and preprocessing	Data preparation
NLTK	Tokenization, stop-word removal and lemmatization	Text preprocessing
Scikit-learn	Training and testing the news classification model	Model development
TF-IDF	Text feature extraction	Feature engineering
Summarization tools	Generating concise news summaries	Summarization

Tool / Software / Equipment	Purpose	Stage Used
Multilingual NLP tools	Translation and multilingual processing	Recommendation
Streamlit	Building the interactive news dashboard	Frontend implementation
Google Colab / Jupyter Notebook	Model development and experimentation	Development & testing
Git / GitHub	Version control and project management	Throughout project

4.3 Test Plan & Verification

The system was tested to verify the functionality and performance of each module. Unit testing was performed for individual modules, followed by integration testing of the complete system.

Table 4.3: Test Cases and Verification Results

Test Case	Input	Expected Output	Status
TC01	News article	Article accepted	Pass
TC02	Raw news text	Cleaned text	Pass
TC03	Preprocessed news	Correct category	Pass

Test Case	Input	Expected Output	Status
TC04	Long news article	Concise summary	Pass
TC05	Selected language	Translated content	Pass
TC06	User preference	Relevant recommendations	Pass
TC07	Complete system	Final dashboard output	Pass

4.4 Results

The developed AI-powered news processing system was tested to evaluate the performance of news categorization, summarization, multilingual processing, recommendation, and dashboard functionality. The testing was performed using the prepared news dataset and the implemented NLP and machine learning modules.

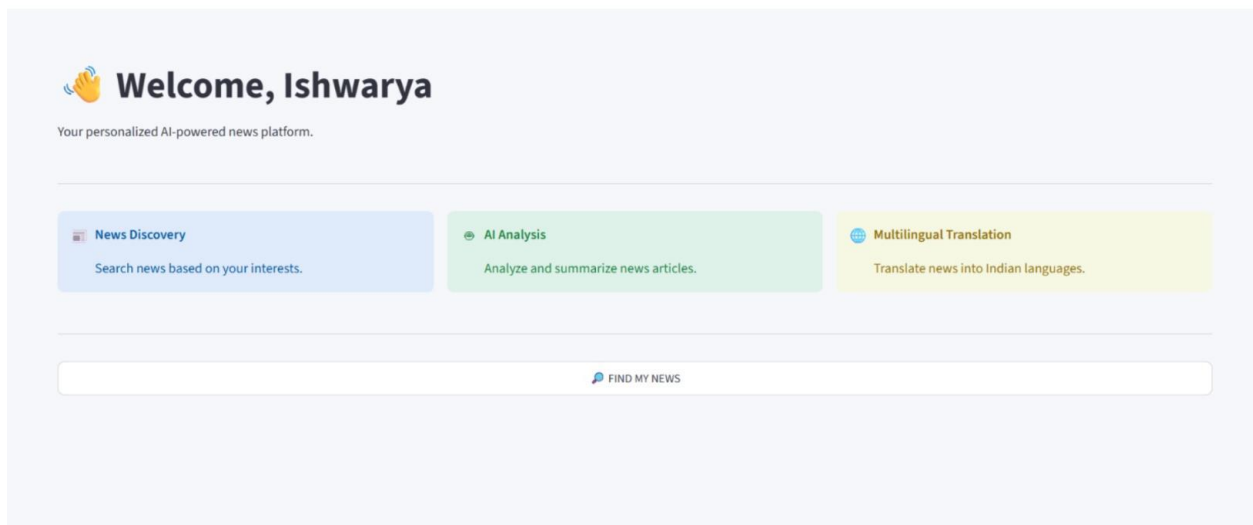


Figure 4.1: System Login and Dashboard Interface

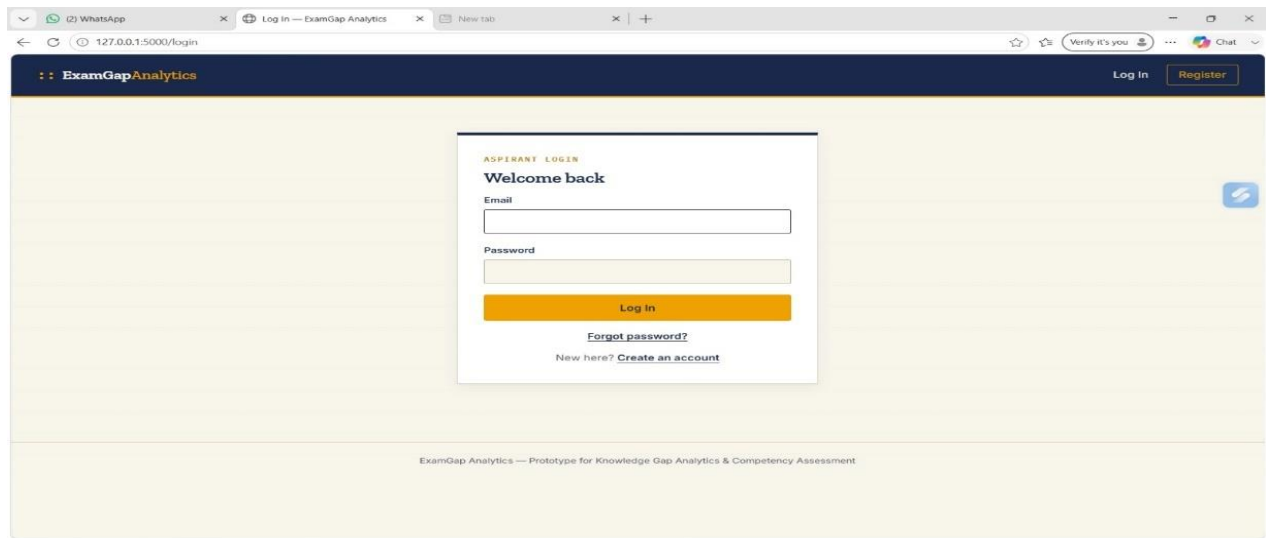


Figure 4.2: User Login Interface

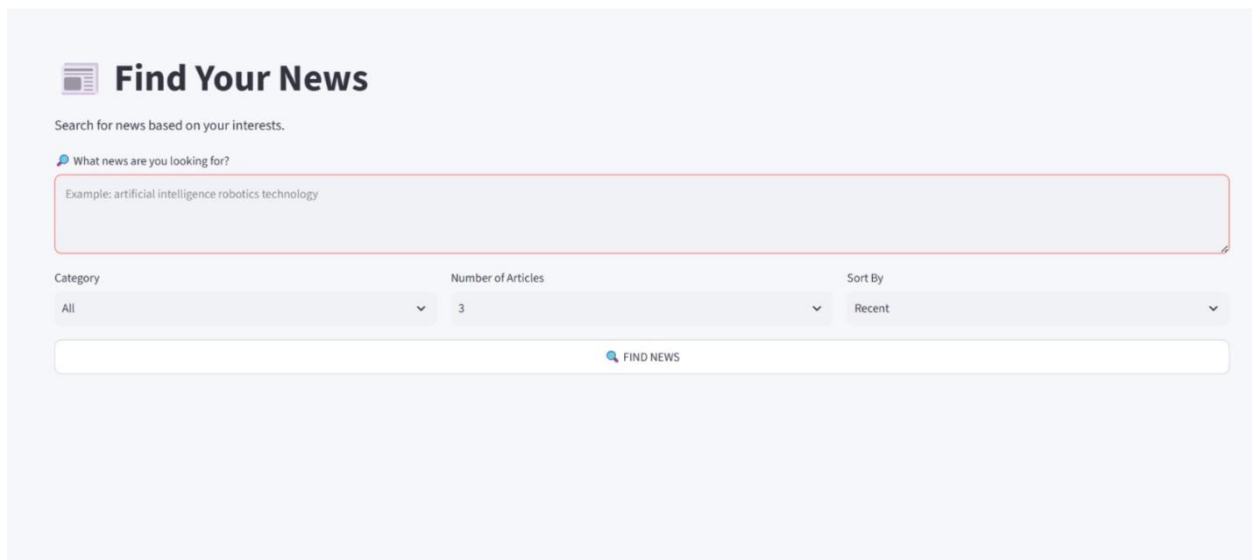


Figure 4.3: News Category Selection Interface

4.5 Data Analysis & Engineering Judgment

The obtained results were compared with the target requirements defined in Chapter 3. Classification accuracy was used to measure news categorization performance, while ROUGE-L

was used to evaluate summarization. Translation latency, recommendation precision, and dashboard response time were used to measure system performance.

Table 4.4: System Parameters and Evaluation Methods

Parameter	Evaluation Method
News Categorization	Accuracy
Summarization	ROUGE-L Score
Translation	Processing Time
Recommendation	Top-5 Precision
Dashboard	Page Load Time
Multilingual Support	Number of Languages

4.6 Comparison with Existing Solutions & Achievement of Objectives

Table 4.5: Comparison with Existing Solutions and Achievement of Objectives

Objective / Feature	Existing Solutions	Proposed System	Achievement
News Categorization	Usually provides basic or separate classification	Automatically classifies news into predefined categories	Achieved

News Summarization	Often available as a separate service	Generates concise summaries within the same system	Achieved
Multilingual Processing	Limited language support in many systems	Supports multilingual news processing and recommendation	Achieved
Personalized Recommendation	Recommendations may be based mainly on general preferences	Recommends relevant news using category, language, and content similarity	Achieved
Integrated News Processing	Different functions are usually handled by separate tools	Combines categorization, summarization, and recommendation in one system	Achieved
User-Friendly Dashboard	May require multiple platforms or interfaces	Provides an integrated dashboard for accessing processed news	Achieved
NLP-Based Automation	Requires manual processing for some tasks	Automates major news processing tasks using NLP and machine learning	Achieved
Overall Project Objective	No single integrated solution for all selected functions	Provides a unified AI-powered news processing system	Achieved

4.7 Strengths & Limitations

Strengths:

- Automatically categorizes news articles into predefined categories.
- Generates concise summaries from lengthy news articles.
- Supports multilingual news processing and recommendation.
- Provides relevant news recommendations based on content, category, and language.
- Integrates categorization, summarization, and recommendation into a single system.
- Provides a simple and user-friendly dashboard.

- Reduces the time required to read and organize large amounts of news.
- The system can be extended with additional NLP and machine learning techniques.

Limitations:

- System performance depends on the quality and size of the dataset.
- Generated summaries may sometimes miss important information.
- Translation quality may vary for different languages.
- Recommendation accuracy depends on available user preferences and news data.
- Advanced NLP models may require higher computational resources.
- The current system may have limited support for languages and news sources.
- Real-time news processing requires additional APIs and infrastructure.
- More testing with larger and diverse datasets is required.

4.8 Project Planning, Roles & Budget

Table 4.6: Project Planning and Activities

Phase	Activity
1	Requirement analysis and identification of project objectives
2	Literature survey and study of existing NLP systems
3	News dataset collection and text preprocessing
4	Development of news categorization module
5	Development of automatic summarization module
6	Development of multilingual recommendation module

7	Integration of all modules with the dashboard
8	Testing, evaluation, documentation, and final presentation

Table 4.7: Team Roles and Responsibilities

Team Member	Responsibility
S. Sahithya	News data collection and text preprocessing
S. Sri Sai Prasanna Durga	News categorization and machine learning model
C. Ishwarya	Automatic news summarization
C. Nithisha	Multilingual recommendation and system integration
All Members	Testing, documentation, presentation, and project review

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The project “AI-Powered News Categorization, Summarization, and Multilingual Recommendation System Using Natural Language Processing” was developed to address the problem of managing the large amount of news available through digital platforms. The proposed system combines Natural Language Processing and machine learning techniques to automate important news processing tasks.

The system consists of four major modules: news data collection and text preprocessing, AI-based news categorization, automatic news summarization, and multilingual news recommendation. The preprocessing module prepares the news content for further processing, while the categorization module classifies articles into suitable categories. The summarization module provides concise versions of lengthy news articles, and the recommendation module helps users discover relevant news based on content, category, and language.

The integration of these modules provides a single platform for efficient news processing and improves the accessibility of information. The testing and evaluation results demonstrate the practical feasibility of the proposed system. Overall, the project shows how NLP and machine learning can be effectively applied to reduce information overload and provide users with relevant and summarized news content.

5.2 Future Work

The system can be further improved and extended in the following areas:

- Integration with real-time news APIs for continuous news collection.
- Addition of more regional and international languages.
- Use of advanced transformer-based models for better summarization and classification.
- Improvement of recommendation using user feedback and personalized profiles.

- Integration of fake-news and misinformation detection.
- Development of a mobile application for easier access.
- Improvement of translation accuracy for low-resource languages.
- Use of hybrid recommendation techniques combining content-based and collaborative filtering.
- Enhancement of system performance for processing large volumes of news in real time.

5.3 Professional Learning & Individual Reflection

This project provided practical experience in Natural Language Processing, machine learning, data preprocessing, model evaluation, and system integration. I learned how to collect and prepare text data, apply NLP preprocessing techniques, develop a news classification model, generate summaries, and implement multilingual recommendation.

Overall, the project strengthened my technical knowledge and provided practical exposure to applying AI and NLP techniques to a real-world problem.

Individual Reflection: C. Nithisha (192424375)

Module 2: AI-Based News Categorization

In Module 2, I Contributed mainly responsible for the multilingual recommendation module and system integration. The AI-Based News Categorization module automatically classifies news articles into predefined categories such as Politics, Sports, Technology, Business, Entertainment, Health, and Science. The module uses NLP techniques to preprocess the news text by removing unnecessary characters, stop words, and other noise. TF-IDF is used to convert the processed text into numerical features, which are then given to a machine-learning classification model to predict the appropriate news category.

My individual contribution was to develop the news categorization process, including text preprocessing, feature extraction, model training, and classification of new articles. I also evaluated the model using accuracy, precision, recall, F1-score, and confusion matrix to verify its performance. This module helps organize large amounts of news automatically and provides categorized content for the subsequent summarization and recommendation modules.



Find Your News

Search for news based on your interests.

 What news are you looking for?

Example: artificial intelligence robotics technology

Category

All

Number of Articles

3

Sort By

Recent

 FIND NEWS

Figure-5.1-News Categorization

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APPENDICES

Appendix A – Detailed Calculations

Random Forest – Competency Classification

$$\text{Accuracy} = \text{Correct Predictions} / \text{Total Predictions} \times 100$$

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

$$\text{F1 Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

$$\text{MAE} = (\sum |\text{Actual Score} - \text{Predicted Score}|) / n$$

$$\text{RMSE} = \sqrt{[(\sum (\text{Actual Score} - \text{Predicted Score})^2) / n]}$$

$$R^2 = 1 - [\sum (\text{Actual} - \text{Predicted})^2 / \sum (\text{Actual} - \text{Mean Actual})^2]$$

Appendix B – Engineering Drawings / Schematics

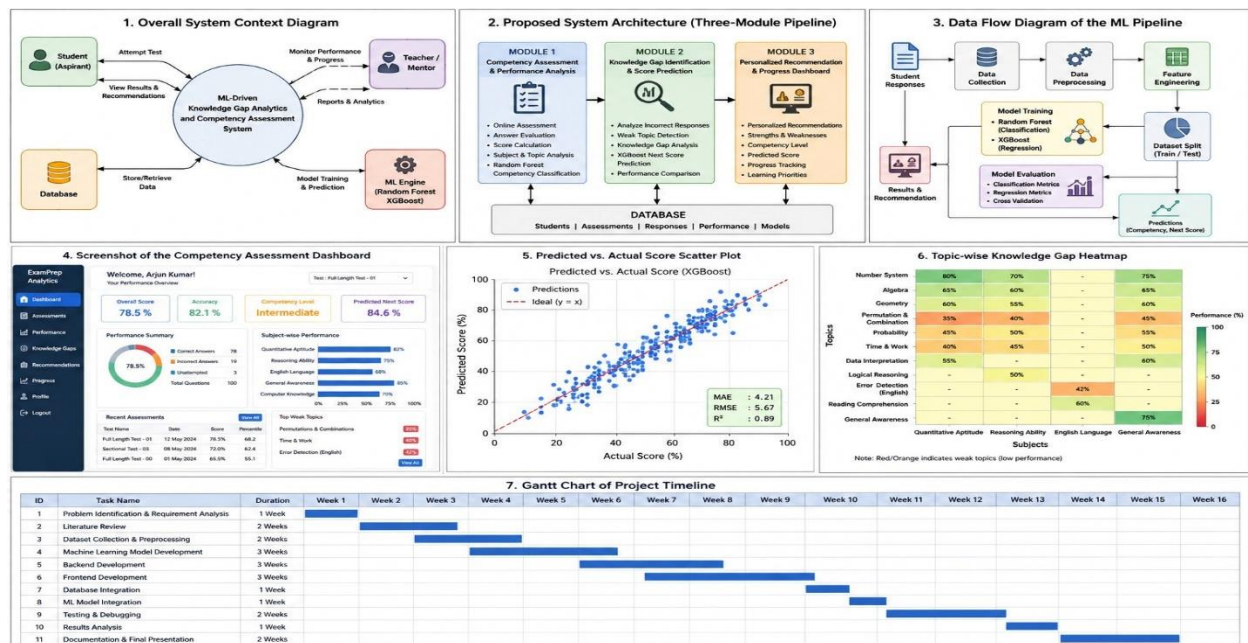
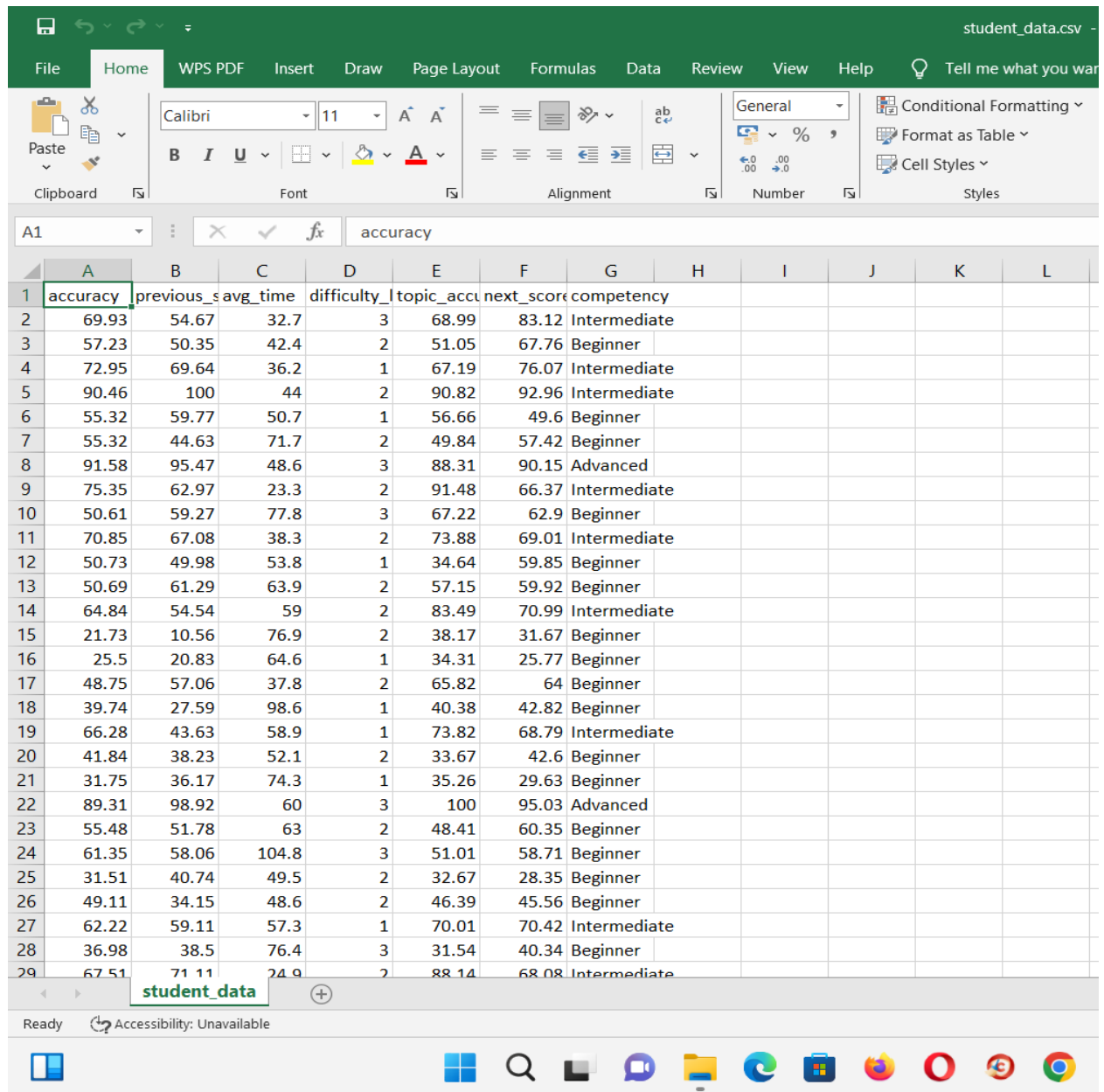


Figure B.1: Engineering Drawings and System Schematics

Appendix C – Source Code / Code Repository Information

<https://github.com/192424239simats-byte/ITA0613/tree/main/Knowledge%20gap%20analytic>

Appendix D – Datasheets



	A	B	C	D	E	F	G	H	I	J	K	L
1	accuracy	previous_avg_time	difficulty	topic_acc	next_score	competency						
2	69.93	54.67	32.7	3	68.99	83.12	Intermediate					
3	57.23	50.35	42.4	2	51.05	67.76	Beginner					
4	72.95	69.64	36.2	1	67.19	76.07	Intermediate					
5	90.46	100	44	2	90.82	92.96	Intermediate					
6	55.32	59.77	50.7	1	56.66	49.6	Beginner					
7	55.32	44.63	71.7	2	49.84	57.42	Beginner					
8	91.58	95.47	48.6	3	88.31	90.15	Advanced					
9	75.35	62.97	23.3	2	91.48	66.37	Intermediate					
10	50.61	59.27	77.8	3	67.22	62.9	Beginner					
11	70.85	67.08	38.3	2	73.88	69.01	Intermediate					
12	50.73	49.98	53.8	1	34.64	59.85	Beginner					
13	50.69	61.29	63.9	2	57.15	59.92	Beginner					
14	64.84	54.54	59	2	83.49	70.99	Intermediate					
15	21.73	10.56	76.9	2	38.17	31.67	Beginner					
16	25.5	20.83	64.6	1	34.31	25.77	Beginner					
17	48.75	57.06	37.8	2	65.82	64	Beginner					
18	39.74	27.59	98.6	1	40.38	42.82	Beginner					
19	66.28	43.63	58.9	1	73.82	68.79	Intermediate					
20	41.84	38.23	52.1	2	33.67	42.6	Beginner					
21	31.75	36.17	74.3	1	35.26	29.63	Beginner					
22	89.31	98.92	60	3	100	95.03	Advanced					
23	55.48	51.78	63	2	48.41	60.35	Beginner					
24	61.35	58.06	104.8	3	51.01	58.71	Beginner					
25	31.51	40.74	49.5	2	32.67	28.35	Beginner					
26	49.11	34.15	48.6	2	46.39	45.56	Beginner					
27	62.22	59.11	57.3	1	70.01	70.42	Intermediate					
28	36.98	38.5	76.4	3	31.54	40.34	Beginner					
29	67.51	71.11	24.9	2	88.14	68.08	Intermediate					

Figure D.1: Student Dataset in Excel Spreadsheet

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)